

Nicolas Rubiano Avila

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TECHNICAL SKILLS:

- **Languages:** Go, Java, Python, SQL, TypeScript
- **Infrastructure:** Kubernetes, Docker, Kafka, RabbitMQ, Redis
- **Databases:** PostgreSQL, MongoDB, Elasticsearch, DynamoDB
- **Frameworks:** Spring Boot, Gin, FastAPI, React
- **Languages (Spoken):** English (Native), Spanish (Native), French (Conversational)

WORK EXPERIENCE:

Backend Developer (*Java, Spring Boot, AWS Lambda, DynamoDB, Postman*)

Sep 2025 - Present

Bloom LLC Delaware, USA (Remote)

- Built serverless authentication system using AWS Lambda and Java — stateless design scales without provisioned servers; session data persisted in DynamoDB for low-latency lookup across thousands of client accounts.
- Verified end-to-end authentication flows via Postman across all API endpoints before each deployment, covering login, token validation, and session tracking for hundreds of active clients.

Teaching Assistant - CPSC416 Distributed Systems (*Python, Go, GitHub*)

Jul 2025 - Sep 2025

University of British Columbia - Vancouver BC

- Debugged and repaired non-functional **Python** grading scripts for **7 labs**, implementing fixes to enable automated evaluation and **GitHub** result publishing for **30+ students**.
- Performed in-depth code reviews of Go-based distributed systems implementations, providing written comments on concurrency patterns, fault tolerance, and code quality

TECHNICAL PROJECTS:

Pairly (*Go, React, Redis, PostgreSQL, MongoDB, Kafka, Kubernetes, Prometheus, OpenTelemetry*)

Feb 2026 - Present

- Engineered real-time collaborative editing across horizontally scaled WebSocket pods via Redis pub/sub — each pod subscribes only to its active room channels, verified at 44 concurrent clients across 3 replicas with 0% error rate.
- Built sandboxed code execution in Docker containers (256MB RAM, 0.5 CPU, no network) via gRPC — 2 services, 14 spans, 127ms p50 end-to-end; Redis SET NX eliminates concurrent execution race conditions per file.
- Implemented point-in-time version history via Kafka event sourcing — 90% of lookups served instantly from MongoDB snapshot cache; full Kafka replay capped at 0.8s for any version across a 30-day edit log.
- Deployed on Kubernetes with independent HPA scaling (3–10 server pods, 2–6 executor pods), 25s/30s graceful shutdown, CI/CD via GitHub Actions, and distributed tracing via OpenTelemetry and Jaeger across 14 spans per execution.

MangaSearch (*Go, Python, FastAPI, Redis, PostgreSQL, Elasticsearch, Docker*)

Jan 2026 - Present

- Engineered a local manga quote search engine — Go owns concurrency (4 goroutine workers BRPOP from Redis, write to PostgreSQL and Elasticsearch at ~16 pages/min), Python owns OCR (containerized FastAPI/EasyOCR); workers auto-recover from transient EOF errors without crashing.
- Designed HashMap diff for filesystem change detection — on rescan of a fully indexed collection, 100% of pages skipped with 0 duplicate jobs entering the queue.
- Delivered fuzzy full-text search via Elasticsearch and Gin REST API — warm queries return in 1–2ms, cold in 62ms; Cobra CLI exposes search, status, and rebuild commands against the running server.

Proximo (*Go, net/http, Zero dependencies*)

Apr 2026 - Present

- Built a reverse proxy from scratch in Go — 5 load balancing algorithms benchmarked under identical real traffic; round robin vs least connections produces 248ms vs 98ms avg latency, a 2.5x difference invisible until measured.
- Implemented 4 rate limiting algorithms (token bucket, sliding window, leaky bucket, concurrent) — token bucket forwarded 16 requests; sliding window forwarded only 5 under identical burst.
- Health checker reroutes away from dead servers in milliseconds — runs off the request path so live traffic never waits; 0 errors recorded during a live backend kill and recovery cycle.

EDUCATION:

Bachelor of Science, Computer Science with Distinction | GPA: 3.7/4.0

Sep 2022 - Nov 2025

University of British Columbia - Vancouver, BC

Associate of Science, General Science with Distinction | GPA: 3.7/4.0

Sep 2020 - Apr 2022

Camosun College - Victoria BC